



storage test

OCZ demonstrates a 1 Terabyte SSD called the Z-Drive that slots into a PCI-Express slot



Desktop Hard Drives

Brand	Hitachi	Seagate	Seagate	Seagate	Seagate	Seagate
Model Name	DeskStar 7K1000.B	Barracuda 7200.11	Barracuda 7200.11	Barracuda 7200.11	Barracuda 7200.11	Barracuda 7200.12
Model Number	HDT721010SLA360	ST31000340AS	ST31500341AS	ST3500320AS	ST3750630AS	ST31000340NS
Price (Rs)	Rs. 14,000	Rs. 5,800	Rs. 8,000	Rs. 3,500	Rs. 5,250	Rs. 6,100
Performance (Out of 100)	64.02	56.31	64.08	63.27	57.84	68.67
Features						
HDD capacity (GB)	931.51	931.51	1397.26	465.76	698.63	931.51
Capacity Per Platter (GB)	334	250	375	250	250	500
Number of Platters	3	4	4	2	3	2
Speed (RPM) / Buffer Size (MB)	7200 / 16	7200 / 32	7200 / 32	7200 / 32	7200 / 32	7200 / 32
Performance						
HDTach 8 MB / 32 MB Zone						
Average Read (MBps)	95 / 94.9	82.8 / NA	108.4 / 107.8	92.6 / 91.8	87.3 / 86.7	106.5 / 106.4
Average Write (MBps)	79.4 / 90.3	62.4 / NA	71.3 / 95.2	NA / NA	NA / NA	NA / NA
CPU Utilisation (%)	2 / 2	2 / NA	2 / 2	2 / 2	2 / 1	2 / 2
Random Access Time (ms)	12.9 / 13.1	12.9 / NA	15.3 / 15.6	14.8 / 14.8	12.6 / 12.6	13.9 / 14
Read Burst Speed (MBps)	262.9 / 262.6	249.9 / NA	247.8 / 251.2	247.2 / 245.8	135 / 134.8	481.2 / 482.6
SiSoft Sandra 2008						
Drive Index, Read (MBps) / Access time (ms)	90.24 / 14	80.12 / 13	102.34 / 15	87.32 / 14	83.21 / 13	100.69 / 14
Drive Index, Write (MBps) / Access time (ms)	68.04 / 6	64.67 / 7	86.59 / 7	NA / NA	NA / NA	NA / NA
Real World Tests (sec)						
4 GB File Write (Sequential / Assorted)	40.3 / 67.9	45.1 / 84	46.6 / 69.9	49 / 72	42 / 84	35 / 65
4 GB File Copy (Seq / Assorted)	137.9 / 164.6	229.8 / 202.2	150.9 / 153.3	159 / 160	225 / 202	143 / 146
Photoshop Image Load (sec)	84.1	64.8	69.7	50	69	65
Farcry level load time (sec)	21.6	21.7	20.4	20	20	21



External Portable Drives



APRIL 2009

Brand	Seagate	Seagate	Western Digital	Western Digital	Western Digital	Western Digital	Western Digital
Model Name	FreeAgent Go	FreeAgent Go	My Passport Elite	My Passport Elite	My Passport For Mac	My Passport Studio	My Passport Studio
Model Number	ST903203FBA2E1	ST905003FAA2E1	WD3200MLZ	WD5000ML	WD5000MEA	WD2000MS	WD5000MT
							
Price (Rs)	Rs. 4,550	Rs. 7,900	Rs. 10,350	Rs. 12,150	Rs. 9,000	Rs. 5,000	Rs. 12,500
Performance (Out of 100)	75.00	77.78	83.43	85.86	83.63	83.73	84.08
Features							
HDD capacity (GB)	298.09	465.76	298.09	465.76	465.76	232.88	465.76
Interface	USB	USB	USB	USB	USB	USB / FW	USB / FW
Performance							
HDTach 8MB / 32MB Zone							
Average Read (MBps)	32 / 31.9	29.9 / 30.2	34.6 / 34.9	34.8 / 35	33.9 / 36.1	34.3 / 34.7	34.8 / 35.8
Average Write (MBps)	26.4 / 30.4	25.6 / 27.1	28.9 / 31.4	28 / 30.8	27.7 / 30.9	27.7 / 30.8	28 / 30.9
CPU Utilisation (%)	5 / 4	1 / 2	5 / 1	4 / 4	1 / 3	5 / 4	4 / 5
Random Access Time (ms)	22.6 / 22.5	21.9 / 21	16.8 / 16.8	17 / 16.8	17.1 / 16.5	17.1 / 17.1	16.8 / 16.5
Read Burst Speed (MBps)	32.9 / 33	32 / 31.9	37.9 / 38	38 / 37.9	36.4 / 37.8	35.2 / 35.5	36.8 / 37.6
SiSoft Sandra 2008							
Drive Index (Read) / Access Time (ms)	30.53 / 20	28.68 / 21	32.51 / 14	32.35 / 13	33.67 / 14	32.29 / 18	31.85 / 13
Drive Index (Write) / Access Time (ms)	30.54 / 12	27.72 / 4	29.44 / 1	29.05 / 1	28.88 / 1	28.77 / 5	29.05 / 1
Real World Tests (sec)							
4GB File Write (Sequential / Assorted)	139 / 168	160 / 164	139.9 / 154.2	134 / 158	139 / 157	134 / 156	139 / 152
4GB File Read (Sequential / Assorted)	157 / 170	154 / 162	161 / 160.9	149 / 162	154 / 166	150 / 160	138 / 167
Internal File Copy (Sequential / Assorted)	454 / 429	338 / 394	303.5 / 323.5	310 / 324	308 / 326	311 / 336	304 / 311
Photoshop Image Load(sec)	92	104	132.2	90	123	94	129